1. Kossoris (1938) Industrial injuries and the business cycle. Monthly Labour Review

Literature Review / Framing / Analysis

The paper has no literature review and frames the question in terms of whether or not industrial injuries increased during the recovery that followed the Great Depression. Other papers often if not always refer to this paper as the foundational paper. Kossoris presents a pro-cyclical view – namely that the economy and worksite injuries (including fatalities) move together.

Model

Kossoris did not use a statistical model. Kossoris seemed to have eyeballed his data.

Core Findings

The paper proposes four hypotheses. The paper also proposes that these arguments can operate in the opposite direction when the economy recovers. They include:

- as the Depression deepened, only the most skilled and seasoned worker kept their jobs, thus reducing injury rates;
- early in the Depression, lay-offs lagged operational reductions; with the lower work tempo, injuries reduced; and the "swollen man-hours" lowered the injury rates[1];
- as the Depression deepened, management turned to only the best and safest equipment, reducing injuries; and
- as the Depression deepened, worked underreported injuries, reducing injury numbers.

The paper finds evidence of hypothesis 1 in the petroleum industry.

The paper also proposes a four-part business cycle hypothesis:

- as employment decreases, injury frequency falls sharply,
- as employment increase again, injury frequency rises sharply as new and former employees come back online,
- subsequent increases in employment produce smaller increases in injury rates, and
- once employment plateaus, injury rate fall as worker become more seasoned.

The paper finds evidence of the first three of this four-part hypothesis in data from the *State of Virginia*. The finds also evidence of the first two of this four-part hypothesis in data from the *National Safety Council.* The paper concludes that the economy-wide and individual industry perspectives do not differ with an exception of four industries. The paper also observes that, compared to the entire number of temporary disabilities, the 1-week disability cases seemed to non-randomly reduce; he concludes that this trend evidences the reporting hypothesis.

[1] Others such as Mouza and Targoutzidis have called this idea the work intensity argument.

2. Steele (1974) Industrial Accidents: An Economic Interpretation. Applied Economics.

Literature Review / Framing / Analysis

Both papers by Mouza and Targoutzidis in 2010 and 2012 refer to this paper. They point to how this article suggests that during upturns the costs of absenteeism or replacement of labor rise and thus occupational accidents decrease. This approached supports a counter-cyclical view. The introduction has a key idea: choosing the right level of safety guards is a truly economic question. Based on the references, this work is based on *UK* data.

He then proposes an analytical model and some related hypotheses. His model is based on an analogy between the machine and the worker. The complete model starts on page 145.

Assumption A: $A = f(M, E)$
$f_e > 0, f_m > 0$
where A = # of breakdowns per period
M = units of maintenance per period
E = factor of production

Assumption B: $f_e > \frac{A}{E}$

Assumption C: $\frac{dP}{dE} = [P_m + P_a f_m] \frac{dM}{dE} + P_a f_e > 0$
where $P = P_m M + P_a A$ = production loss
P_a = production loss per breakdown
P_m = production loss per unit maintenance

Assumption D: $\frac{dP_a}{d\frac{V}{U}} > 0$
where V = unfilled vacancies
U = employment level

Assumption E: $\frac{d\frac{V}{U}}{dE} > 0$

Hypothesis 1: "the optimal expansion path [is] such that the accident rate remains constant as employment increases". In other words, $\frac{df(M,E)}{dE} = \frac{A}{E}$.

Hypothesis 2: given "the condition for minimizing production loss..., accidents would be expected to vary inversely with the cost involved". In other words, $\frac{\partial A}{\partial P_a} < 0$ and thus $\frac{\partial A}{\partial \frac{V}{U}} < 0$.

Hypothesis 3: accidents and employment are inversely related. In other words, $\frac{dA}{dE} < 0$.

Hypothesis 4: more accidents come with more overtime (T). In other words, $\frac{\partial A}{\partial T} > 0$.

Model
The outcome variable for model 1 is reported accidents at factory process p and time t. The exogenous variables for model 1 include a labor scarcity variable, described below, hours (in 1000s) of overtime work, employment in industry y at time t, and a dummy for the third quarter season. The labor scarcity variable is the quotient of unfilled vacancies in industry y at time t and the number employed at time t.

Model 2 is the same but omits the labor scarcity variable. Model 3 is the same as model 1 but adds a quarterly time variable.

The outcome variable on model 4a is the quotient of the accidents in region r over the employment level in region r. The exogenous variables for model 4 include the quotient of unfilled vacancies in region r over number of employed in region r, a hypothetical accident rate, and employment in region r. The hypothetical accident rate is "the number of accidents which would have occurred in region r if each of its industrial sectors had experienced the national rate relevant to those sectors" divided by the employment in region r. Model 4b is the same but omits employment in region r.

Core Findings
Regression model 1 affirms hypotheses 1, 2, and 4. Model 2 affirms hypothesis 3. He also runs the model 1 regressions for nine disaggregated industries: food, chemicals, metal, manufacturing, textiles, clothing, bricks, timber, and paper. Timber was the only anomalous industry in that the labor scarcity variable was positive. He also standardized the difference between marginal accident rates (from regression model 1) and average accident rates and found the difference significant at the 10 percent level.

He brings up another question: if "legislation or propaganda are producing a distinct trend it is arguable that any of the time-series variable…might be acting as a proxy from time trend." So he adds a quarterly time variable (i.e. model 3). The other coefficients do not change as a result of this additional, suggesting that they were not acting as such proxies.

He then creates a cross-sectional model (model 4) and finds that 1) accidents do not vary with employment, 2) industries with scarce labor worked the least overtime per worker, 3) accidents and labor scarcity related inversely, and 4) the accident rate varies with a hypothetical accident rate. However the coefficient on the hypothetical rate was higher than expected.

Methodology
The models are determined reasonably and no statistical manipulations were needed.

3. Shuster and Rhodes (1985) The Impact of Overtime Work on Industrial Accident Rates. Industrial Relations.

Literature Review / Framing / Analysis
This paper is not referred to often, is dated, and is perhaps not a perfect journal but it is very much on point. This paper focuses on the notion of fatigue, which relates well the larger concept of how booms may produce increased injuries. These authors defined fatigue as "boredom" and relate it to reduced motivation. This work is based on American data.

They present 6 hypotheses:

1) a positive relationship exists between accident severity and overtime worked during an uninterrupted span of pre-accident days,
2) a person who has had an workplace accident will more likely have worked overtime preceding the accident than preceding a non-accident day,
3) a person who has had an workplace accident will have worked a greater number of overtime minutes preceding the accident than preceding a non-accident day,
4) a person who has had an workplace accident will have worked a greater number of overtime days preceding the accident than preceding a non-accident day,
5) the influence of overtime on accident incidence and severity is greater the closer the proximity the overtime is to the accident, and
6) older workers exhibit greater accident severity and incidence.

Models
The outcome variable for model 1a and 1b was accident incidence (i.e. occurrence of injury regardless of time lost) and the log of accident incidence, respectively. All these models include controls for gender, job and organizational tenure, whether the overtime was voluntary, and whether the work was self-paced or machine-paced. The other exogenous variable was overtime preceding the accident (i.e. any overtime day occurring during a seven-day period preceding the accident).

The outcome variable for model 2 was accident severity (i.e. the number of work days lost). The exogenous variable was overtime minutes preceding the accident (any overtime minutes occurring during a seven-day period preceding the accident). Model 2b added age as an exogenous variable. Model 2c added the product of age and accident severity as an exogenous variable – a so-called moderating variable.

Core Findings
The results suggest that overtime work is related to accidents and accident severity. For those having a lost-time accident, more consecutive days of overtime with more minutes of overtime meant more severity of accident. They affirm all hypotheses but 5 and 6.

Methodology
"The distribution of a variable was found to be nonnormal, the data were transformed to approximate a normal distribution." Otherwise, the models speak for themselves.

4. Nichols (1991) Labour Intensification, Work Injuries, and the Measurement of Percentage Utilization of Labour (PUL). British Journal of Industrial Relations.

Literature Review / Framing / Analysis
Both papers by Mouza and Targoutzidis in 2010 and 2012 refer to this paper. They point to how this paper suggests that the pro-cyclical behavior of accident rates really reflects a lag between economic downturns and unemployment – not a true relationship between economic downturns and accidents. Key here is that this idea does not operate on the macro/lifestyle-related avenue. This work is based on British data.

Model
He ran ten models. The first six have fatality rate as the outcome variable. The exogenous variables for all these models include total weekly hours worked, the engagement rate (which he does not define), and fixed investment lagged by one year. Model 1 covers 1960 to 1985. Model 2 covers 1971 to 1985. Model 3 covers 1971 to 1985 and adds PUL. Model 4 covers 1971 to 1985 and adds output per hour only (as in it does not include PUL). Model 5 covers 1972 to 1985 and adds the delta of PUL only. Model 6 covers 1972 to 1985 and adds the delta of output per hour only.

The second four models have the rate of change of the fatality rate as the outcome variable and cover 1972 to 1985. The exogenous variables for these models include the delta of total weekly hours worked, the delta of the engagement rate, and the delta of fixed investment lagged by one year. Model 8 adds the delta of PUL only. Model 9 adds the delta of output per hour only. Model 10 adds only the delta of output per hour and a dummy for whether or not the *Health and Safety at Work Act* was in effect.

Core Findings
"A measure called the percentage utilization of labour (PUL) claims to monitor the intensity of labour in British manufacturing...This paper introduces the PUL measure and indicates why it might be thought particularly likely to register an increase in the 1980s. It then examines variation in PUL and in industrial injury rates in British manufacturing, both in the early part of the 1980s and over a longer period, in order to assess whether changes in PUL have tended to be associated with changes in safety. No consistent relationship is found."

He argues that Britain experience a reduction in the 'porosity of labour' during the 1980's and shows a figure from 1971 to 1990 of generally rising PUL (presumably for British manufacturing). By reduced porosity, I took it that he felt people worked harder.

He then examines to what extent PUL correlates with two measures: a major injury rate and a fatality rate. Regarding the former, he says, "there is no evidence" that it and PUL correlate well. His most demanding method is eyeballing industry specific figures for Mechanical Engineering, Metal Goods, Food, Drink and Tobacco, Motors and Parts, Footwear and Clothing and Chemicals. (He notes that the major injury rate may itself not measure work intensity properly, confounding the question yet further.) Regarding the

fatality rate, his most demanding method is multivariate regression. I presented his model above. His finding is that "regression analysis does nothing to dispel doubt about PUL's capacity to make a distinct contribution to the analysis of industrial safety, or to confirm its uniqueness, when compared with output, as a measure of labour intensity."[2]

Methodology
The models are determined reasonably and little additional statistical manipulations were needed.

[2] Regarding PUL, it "was developed by Smith-Gavine and Bennett. It is constructed from data supplied from factory records on a monthly basis."

"A key objective…was that the index should serve as a means to separate out that component of labour productivity that could properly be regarded as what they term 'technological productivity'."

"The index was intended to serve as a measure of 'economic temperature' and was based on the assumption that output is in part a function of the throughput of work (the volume of work put behind an operative's machine)…PUL is intended to monitor changes in the intensity of labour, whether workers have been driven to such intensity through fear of losing their jobs, whether it has come about by the 'pores' in the working day having been closed by some other mechanism, or even whether workers…enjoy it. PUL derives from factory records which purportedly measure how far workers actually achieve the standard minutes and seconds allocated to jobs by work measurement techniques."

5. Saloniemi and Oksanen (1998) Accidents and Fatal Accidents – Some Paradoxes. Safety Science.

Literature Review / Framing / Analysis
The paper focuses on worksite injuries in construction and manufacturing industries (each separately) in Finland from 1977 to 1991. Keep in mind only Mouza and Targoutzidis refer to this work. They say on page 61 that "linear regression" was used. However, in construction and labeling, the tables reflect the use of correlation as the analysis tool. They may be using the term regression in a wide sense. This work is based on Finnish data.

Model
The authors produced four correlations, two for construction and manufacturing each. In other words they measure how the "fatality rate" (i.e. fatal accidents per 100 000 persons employed) and "accident frequency" (i.e. accidents per million working hours) related to each other and four additional variables: cubic meters under construction, the number of employees, working hours, and unemployment rate. For obvious reasons, they do not include cubic meters under construction in the manufacturing analysis.

Core Findings
"The correlations between the fatality rate and the possible explanatory variables remain weak throughout. In the manufacturing industry none of the correlations reached statistical significance. In building and construction the variables describing average labour market situations…show only a minimal or no correlation with the fatality rate."

Regarding construction, the paper proffers an inverse relationship between the fatality rate and both cubic meters under construction and accident frequency. The paper then focuses further on construction – observing that i) cubic meters under construction and the fatality rate relate inversely and ii) cubic meters under construction and accident frequency move together, which suggests that if "an increase in cubic metres means more accidents, it also means fewer fatal accidents".

In their discussion they write:

> The analysis showed that there is no meaningful connection between fatal accidents and business cycles. None of the labour market indicators (number of workers and hours, unemployment rate) in either the manufacturing or construction industries were associated with deaths at work. In this sense the results differ clearly from most earlier findings on non-fatal accidents. In terms of fatal accidents the present data tend not to support the hypothesis of a pro-cyclic relation between accidents and economic activities.
>
> In the manufacturing industry this lack of correlation with cycles applied across the board, but in building and construction there was one notable exception. Here, declining production (measured in cubic metres under construction) was followed by an increase in the fatality rate. The accident frequency declines under the same conditions, which leads to the question of causation.

> In the case of the construction industry, the present material does not corroborate the hypothesis of the iceberg-like constitution of safety problems.

An “iceberg-like constitution of safety problems” suggests that a single fatality comes with and *in a sense* represents a larger number of non-fatal accidents.

Methodology
The models are determined reasonably and little additional statistical manipulations were needed.

6. Gerdtham and Johannesson (2003) A Note on the Effect of Unemployment on Mortality. Journal of Health Economics.

Literature Review / Framing / Analysis
This paper is largely a macro/lifestyle-related mortality paper. They do not report any findings specifically related to worksite injuries or accidents except within a larger disaggregated category titled "Other external". They mention as a matter of theory not empirics that greater unemployment may decrease the risk of job related accidents. They refer to "Other external" and include in it "motor vehicle accidents, accidents and homicides". (In some sense I did not need to look at this article.) This work is based on Swedish data.

Model
The authors use a probit model. The outcome variable is the survival status of the respondent at the end of the follow-up, coded 1 if dead and 0 if alive. The exogenous variables included a dummy for unemployment, a categorical variable for self-reported health (including poor, fair, and good), a categorical variable for physical functionality (involving running and stair climbing), a dummy for hypertension, and a dummy for whether or not respondents' parents were living. Additional exogenous variables included annual disposable income, and the annuity of net wealth age (at the inclusion in the study), gender, immigration status, education, marital status, number of children, geographical location (county in Sweden), and level of urbanization.

Core Findings
"Unemployment significantly increases the risk of being dead at the end of follow-up by nearly 50% (from 5.36 to 7.83%). In an analysis of cause-specific mortality, we find that unemployment significantly increases the risk of suicides and the risk of dying from "other diseases" (all diseases except cancer and cardiovascular), but has no significant effect on cancer mortality, cardiovascular mortality or deaths due to "other external causes" (motor vehicle accidents, accidents and homicides)." The coefficient on "other external causes" is negative but not significant.

Methodology
The models are determined reasonably and little additional statistical manipulations were needed.

7. Terres de Ercilla et al. (2004) Economic fluctuations affecting occupational safety. The Spanish case. Occupational Ergonomics.

Literature Review / Framing / Analysis

Both papers by Mouza and Targoutzidis in 2010 and 2012 refer to this paper. This paper introduces two views of understanding: Kossoris's intensification view and Nichols' vulnerability view[3]. Intensification is simple: more work, more work intensity, and thus more injury. It is a pro-cyclical view. The vulnerability view suggests that during upturns labor has power to demand greater safety benefits, which ensures fewer injuries; and during downturns, visa versa. It is a counter-cyclical view.

The author then proposes an integrated view, which borrows from Kossoris and Nichols and others. In the integrated view, they propose a real effect and a reporting effect. The real is negative (i.e. counter-cyclical). It stems from the thinking that the Nichols view trumps the Kossoris view; they specifically recall Ruhm to claim that the counter-cyclical dominates. The reporting view stems from the Boone and van Ours[4] thesis: in upturns people feel more comfortable reporting less severe injuries (i.e. pro-cyclical).

They propose five hypotheses:

1) there is a negative correlation between incidence indexes (i.e. IIT) and unemployment rates given the policy controls,
2) there is a negative correlation between incidence indexes of major accidents (IIM) and unemployment rates given the policy controls,
3) there is a negative correlation between incidence indexes of fatal accidents (IIF) and unemployment rates given the policy controls,
4) (it has two parts)
    a. accident reporting – defined as the quotient of the global incidence index over the major or fatal accident index (i.e. TtoM or TtoF) – has changed substantially over the test period,
    b. major accident reporting – defined as the quotient of the incidence index of major accidents over that of fatal accidents (i.e. MtoF) – has remained relatively constant over the test period,
5) accident reporting – defined as the quotient of the global incidence index over the fatal accident index (i.e. TtoF) – is negatively related to the unemployment rate given the policy controls.

Model

For correlations, they examine the natural log of certain incidence indexes, the natural log of the unemployment rates in percentages, the natural log of certain instrumental series of

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[3] Nichols does not present this thesis in Nichols (1991) that I summarized here. He does so in yet another paper, which I just found on October 6th. I had just received the Terres de Ercilla et al. (2004) this week. I will have mentioned this in my email.

[4] You may be wondering why they refer to a paper that came out in 2006. They read the 2002 university version, which was published and become Boone and van Ours (2006).

severity, a "time tendency in annual data for the period" of 1988 to 2002, and a policy proxy: the natural log of the number of legal regulations existing at year end.

The incidence indexes as in accidents per 100,000 workers include: IIT (number of accidents with absence from work), IIM (major accidents), and IIF (fatal accidents). The series of severity include: TtoM (number of accidents divided by major accidents), TtoF (number of accidents divided by fatal accidents), and MtoF (number of major accidents divided by fatal accidents). My understanding is that IIT is the "global" incidence index.

For regressions, they run six models. The outcome variables for each model include: IIT, IIM, IIF, TtoM, TtoF, and MtoF. The exogenous variable for all these models include: the natural log of the unemployment rates in percentages, the policy proxy, and (presumably) a time tendency variable.

Core Findings

Under correlation, they affirm hypotheses 1 and 5. They seem to affirm hypothesis 4 visually. They also note (again visually) a clear pro-cyclical relationship given fluctuations of global incidence index and unemployment rates.

Under regression, they affirm hypotheses 1, 2, and 3. They do not affirm hypothesis 5 although the sign was as expected. They, however, notice that the policy variable was not significant with the general incidence index and was with the major and fatal incidence indices, which was odd to them. Also, the unemployment rate and TtoM relate negatively. The policy variable and all three of the quotient variables – TtoM, TtoF, and MtoF – relate positively each in separate regressions.

Methodology

The models are well determined and little additional statistical manipulations were needed.

8. Boone and van Ours (2006) Are recessions good for workplace safety. Journal of Health Economics.

Literature Review / Framing / Analysis

It focuses on absenteeism and reporting and presents a clear alternative but still pro-cyclical thesis. They include a theory section. In this theoretical model, each worker has an exogenous probability of experiencing an accident; the utility damage of each accident has a distribution; after an accident, the worker decides to report and get compensation. Compensations grow iteratively but worker stigma can set in, subjecting the worker to a potential layoff or implicit discouragement by the firm. This work is based on data from 16 OECD nations.

Models

The paper presents five primary multivariate, fixed effect regression models for non-fatal and fatal injuries each – ten models altogether. The outcome variable in the first five models is number of non-fatal accidents as a percentage of the number of workers; for the second five models it's fatal accidents as a percentage of the number of workers. Note: all models have "country fixed effects, calendar year fixed effects, country-specific time trends and dummy variables for structural breaks". They used dummy variables to address missing-data problems for "structural breaks" as they call them.

Model 1 includes the natural log of the unemployment rate. Model 2 includes the log of the unemployment rate and the first difference of the log of the unemployment rate.[5] Model 3 includes the log of the unemployment rate and the first difference of the log of the employment rate. Model 4 includes the log of the unemployment rate and the log of the average number of hours of work. Model 5 includes the log of the unemployment rate and employment share of manufacturing and construction. Model 6 to 10 have the same respective right-hand sides.

They introduce another set of models. The outcome variable is, as before, non-fatal accidents as a percentage of the number of workers. But they do not do a set of fatal accidents models in this case. The exogenous variables include the log of the unemployment rate and log of fatal accidents as a percentage of the number of workers. The fatal accidents variable works as a proxy for an accident probability. Then, they run several different regressions using these two exogenous variables with the same controls. The regressions include a full run, a 1976 to 1989 run, 1990 to 2001 run, an EU run, four industry-specific runs, and different institutional configurations.

The authors introduce one last model. The outcome variable for this model (presumably, though they hardly make it clear) is non-fatal accidents. The exogenous variables include the log of the unemployment rate, fatal accidents (again as a proxy for an accident probability), two interactions, and the usual controls; the interactions include: one

[5] I was not sure about the order of operation on the latter variable. I wrote it consistent with their notation: $\Delta \ln (u_{i,t})$. But I suspect the difference was taken first. So the way I wrote it above it not correct. This same observation applies later first difference variables.

between the log of the unemployment rate and the demeaned unemployment benefits level and the other between the log of the unemployment rate and a dummy for whether or not the country collects reported accident data. (I presume they used the dummies for "structural breaks" as with the earlier models.)

Core Findings

The authors state their conclusion early on regarding their core models, which I call, models 1 to 10:

> the unemployment rate is negatively related to the workplace accident rate while working hours and the change in the employment rate have no effect on the workplace accident rate. Furthermore, cyclicality is found only for non-fatal workplace accidents. From this evidence and additional sensitivity analyses we conclude that the procyclical behavior of workplace accidents is most likely related to reporting rates and not to actual fluctuations in accidents.

The authors again summarize their results quite well:

> All in all, it seems that high unemployment rates have a significant negative effect on workplace accident rates whereas changes in unemployment, changes in employment, working hours and industrial structure do not affect non-fatal accident rates. Fatal accident rates are not influenced by labor market conditions at all. From this we conclude that cyclical fluctuations in the non-fatal workplace accident rate are driven by the reporting behavior of the workers and not by changes in workplace safety.

The author presented another set of models, which address among other things certain industries and institutional configurations. I do not exhaustively outline the results from this set of models. The results, which are on page 19 to 20, are not very interesting with the two following exceptions. Based on these models, the authors address the question of institutions and non-fatal accident rates – writing:

> First, we make a distinction between countries with a high union density and countries with a low union density. We find that there is hardly any difference in parameter estimates for the two groups of countries. Then, we make a distinction between countries with low unemployment benefits and countries with high unemployment benefits. Now, we do find a difference. [F]or countries with low unemployment benefits the cyclicality in workplace accidents is stronger than for countries with high unemployment benefits. This confirms the reporting behavior explanation since there is no effect of unemployment benefits on workplace conditions.

They introduce another model, which I outlined above, and observe: "as expected in countries with low unemployment benefits and countries that have reported accident data the effect of unemployment on non-fatal workplace accidents is stronger than in other countries".

Methodology

The models are well determined and little additional statistical manipulations were needed. The authors share a sensitivity analysis, which corroborates their findings.

9. Gerdtham and Ruhm (2006) Deaths rise in good economic times: evidence from the OECD. Economics and Human Biology

Literature Review / Framing / Analysis
This macro/lifestyle-related paper barely addresses worksite injuries as an empirical question and does so only among other forms of mortality and within a disaggregated "other accidents" category. The paper is pro-cyclical.

Model
The paper presents three sets of multivariate, fixed effect regression models, which I will call model 1a to 1f, model 2a to 2h, and model 3a and 3b.

The outcome variable of model 1 is the log of the mortality rate. The exogenous variables of model 1a include national unemployment rate, a year-specific intercept, a country fixed-effect, "country dummy variables and controls for the percentage of the country population who are: men, 15–64 years old, 65–74 years old, and 75 years and over", year dummy variables, and country-specific time trends. Model 1b adds the log of per capita income. Model 1c drops the log of per capita income and omits country-specific time trends. Model 1d also drops the log of per capita income and omits year effects. Model 1e drops the log of per capita income and substitutes "the (population-weighted) mean OECD unemployment rate for the national rates". Model 1f drops the log of per capita income and includes both the OECD and national unemployment rates.

Model 2a and 2b are literally the same as model 1a and 1b. The outcome variable of model 2c and 2d is the mortality rate (as opposed to its log). Models 2e and 2f are for the 10 largest economies. Models 2g and 2h focus on 1983 to 1997. Models 2i and 2j focus on Canada, Japan, the UK and the US.

The outcome variable of model 3 is the log of the mortality rate by cause: cancer, major cardiovascular disease, influenza/pneumonia, liver disease/cirrhosis, auto accidents, other accidents, suicides, homicides, and infant deaths.

Core Findings
"The main finding is that total mortality and deaths from several common causes rise when labor markets strengthen. For instance…a 1% point decrease in the national unemployment rate is associated with growth of 0.4% in total mortality and the following increases in cause-specific mortality: 0.4% for cardiovascular disease, 1.1% for influenza/pneumonia, 1.8% for liver disease, 2.1% for motor vehicle deaths, and 0.8% for other accidents. These effects are particularly pronounced for countries with weak social insurance systems, as proxied by public social expenditure as a share of GDP." The term "other accidents" includes worksite injuries.

Methodology
The models are well determined and little additional statistical manipulations were needed.

10. Song, He, and Li (2010) Longitudinal Relationship between Economic Development and Occupational Accidents in China. Accident Analysis and Prevention

Literature Review / Framing / Analysis
This paper approaches the subject with a very wide focus, addressing expansions as well as long-term development. The paper is pro-cyclical.

Model
The authors indicate that they used regression. To my knowledge, they do not actually present a regression model. They present statistical testing models related to series stability, Granger causality, impulse response analysis, and cointegration.

Core Findings
The findings indicated no relationship between occupational accidents and economic scale from 1953 to 1978. However, fatality rate per 100,000 workers was a "conductive" variable to gross domestic product per capita from 1979 to 2008. Furthermore, economic cycle was an indicator to occupational accidents during from 1979 to 2008. Variation within economic speed also had influenced occupational accidents in short term.

Methodology
As far as I can tell, their methodology is not even sophisticated econometrics. It arises from a non-economic field.

11. Mouza and Targoutzidis (2010) The Effect of the Economic Cycle on the Workplace Accidents in Six European Countries. Ege Academic Review

Literature Review / Framing / Analysis
This literature review is informative; and the paper focuses on the workplace angle. This paper is generally counter-cyclical.

Model
The author uses a core multivariate regression model using panel data for fatal and non-fatal accidents. They created models for each of Finland, Italy, France, Switzerland, Austria, and Germany. However, for non-fatal accidents, they have no Austria model. So, they create 11 models in total. The core outcome variable, for each set of models, is fatal and non-fatal injuries per 100,000 workers. The exogenous variables include unemployed person per 100,000 workers[6], GDP per capita, and average cost of labor per unit of output. They do not speak of other controls, which seems antiquated. They argued that because they used multiple nations, they effectively controlled for various factors.

Core Findings
The paper "indicates a counter-cyclical behavior of workplace accidents. Output per capita is the only parameter whose impact is statistically significant in all cases." That is pretty much what the paper delivers. Delving a bit deeper we see a bit more on their analysis.

For three countries, Germany, Austria, Switzerland, the authors report negative correlation between unemployment and accident rates (both fatal and non- fatal) while for Italy they report a positive correlation; and for France and Finland they report no significant correlation.

Unit labor cost is positively correlated with fatal accidents (although in some cases this correlation is not significant); for Germany it is statistically significant. Unit labor cost is positively correlated with non-fatal accidents; only for Italy and Switzerland is it statistically significant; the others are either not significant. The unit labor cost findings are actually a sign of pro-cyclicality but they fail to note that; they actually ignore the Germany finding perhaps given the coefficient's size.

Unemployment has no significant effect on fatal injuries; and, even for non-fatal injuries this effect is significant in only two cases: Finland and France.

Methodology
The models are simple and little additional statistical manipulations were needed.

[6] The per 100,000 workers was not made explicit in paper.

12. Svensson (2010) Economic Upturns are good for Your Health but Watch Out for Accidents: A Study on Swedish Regional Data 1976 – 2005. Applied Economics

Literature Review / Framing / Analysis
This paper addresses worksite injuries in their cause-specific breakdown of mortality. Worksite injuries are not disaggregated from its cause-specific category: "Other Accidents". This is a pro-cyclical paper in terms "Other Accidents".

Model
The author uses a multivariate regression model with fixed effects and interactions using panel data. The outcome variable is log mortality rate in a given region and time; the results include total mortality, male mortality, and female mortality models. The exogenous variables include a year-specific term, a region-specific term, a regional specific time trend term, a regional unemployment term, and x vector of covariates.

Core Findings
The abstract reads as follow:

> Overall mortality is unrelated to changes in the unemployment rate, while the biggest cause of death (heart disease) decreases when the unemployment rate decreases. At the same time, other accidents, including job-related accidents, increases when the unemployment rate decreases.

To borrow from the introduction:

> The analysis in this article reveals that no robust relationship can be found between the total, age or gender-specific mortality and the business cycle. The 10 specific causes of death indicate that this is reasonable, considering that the effects are different for different causes. Two robust results are found in the cause-specific analysis. Primarily Other Accidents, including job-related fatalities, significantly increase in economic upturns, while Heart Disease mortality, the most common cause of death in Sweden as well as in many other developed countries, tends to decrease in economic upturns.

Methodology
The models are well determined and little additional statistical manipulations were needed.

13. Boone et al. (2011) Recessions are bad for Workplace Safety. Journal of Health Economics

Literature Review / Framing / Analysis

This paper again focuses on the reporting thesis from Boone and vans Ours (2006). In the earlier paper, reporting an accident lead some form of compensation. In this paper, reporting an accident leads to investments to prevent accidents. This paper allows for an analysis of over-reporting not just under-reporting. The authors also include a theory section with positive analysis and welfare analysis.

In the positive analysis, we have two periods. In period one, an accident might have happened and the workers must decide to report or not and the firm must invest in safety and alter its perception of the worker. The likelihood of lost demand also comes into the mind of the firm and the worker. Proposition 1: as the likelihood of lost demand rises, reporting likely falls; as the expected damage of an accident rises, the likelihood of reporting also rises and the impact on likelihood of lost demand and likelihood of an accident both attenuate.

In the welfare analysis,

> "there are two main imperfections that cause the equilibrium… [in the model] to deviate from the socially optimal outcome [investment in safety measures]. First, both the worker and the firm only consider their own payoffs not the sum of payoffs… Second, after an accident the worker learns his own…[accident probability] but cannot credibly reveal this to the employer. Hence the firm bases its investment decision on… [an estimate of that accident probability] while it is socially optimal to base this decision on [the true accident probability]. Related to this, the [possible] decision to fire a worker…after an accident should also be based on [the true accident probability] and not on the accident itself. If the worker has [less than expected] accident probability the social planner prefers the workplace accommodating investment to firing the worker. However, when [the accident probability is more than expected] even the social planner prefers to fire a worker rather than investing in prevention of a further accident.

Proposition 3[7] says that there exist an ideal likelihood of lost demand above which the firm will under invest in safety and below which the firm will over invest.

Model

The paper presents two core multivariate regression models with fixed effects models using panel data. Model 1 tests the idea that reporters get fired; also, they performed this model on moderate and severe accidents. Model 2 tests the relationship between reporting and the economy; also, they performed this model on moderate and severe accidents; and, given the model determination, they report results at the firm level and at the individual level.

[7] There is no proposition 2 for some reason.

The outcome variable on model 1 is a binary variable for whether or not a worker is fired in period t + 1. The exogenous variables include current and past accidents of worker i as reported to firm j at time t, control variables, a worker fixed-effect, a firm fixed-effect, a calendar month dummy, and a year dummy. (All exogenous variables are at time t except the worker fixed-effect, which is not time indexed.) Control variables include "a constant, the state unemployment rate for male, blue-collar workers, the sickness rate of the last 2 years of worker i in firm j (and its square), the daily wage (in logs), age (and its square), tenure (and its square), and experience (and its square)."

The outcome variable on model 2 is a binary variable for whether or not a worker reports an accident at time t. The exogenous variables include the firing rate of the firm, a worker fixed-effect, a firm fixed-effect, a calendar month dummy, a year dummy, and control variables. (All exogenous variables are at time t.) Control variables include "a constant, the sickness rate of the last 2 years of worker i in firm j (and its square), the daily wage (in logs), age (and its square), tenure (and its square), and experience (and its square)."

Core Findings

The paper finds that "workers who reported an accident in a particular period of time are more likely to be fired later on" and that "recessions influence the reporting of moderate workplace accidents: if workers think the probability of dismissals at the firm level is high, they are less likely to report a moderate workplace accident."

Methodology

The models are well determined and little additional statistical manipulations were needed.

14. Mouza and Targoutzidis (2012) The impact of the economic cycle on fatal injuries. The case of UK 1971 – 2007. Quality and Quantity

Literature Review / Framing / Analysis
The authors build on the view of Terres de Ercilla et al and investigate the idea that different economic variables will have different relationships. The paper coins the term 'flexible' workers to mean temporary, part-time, and self-employed workers – all of who face greater risks. As such, they address hours worked in their model.

Model
The authors use a multivariate regression model. The outcome variable is fatal injury levels (not rates) in year i. The explanatory variables include GDP per capita (in $1,000s) at year i – 1, the unemployment rate in year i – 1, average hours actually worked in year I, and a dummy set to 2 if after 1985 and 1 if not.

Core Findings
The paper investigates the effect of the economic cycle on fatal workplace accidents in the United Kingdom from 1971 to 2007. The paper finds

- elastic and positive relationship between working hours and fatalities,
- inelastic and negative relationship between GDP per capita and fatalities, and
- inelastic and positive relationship between unemployment rate and fatalities.

These findings affirm Kossoris's second factor – namely work intensity. These findings do not affirm Steele's view that labor become scare during upturns and thus absenteeism and replacement become more expensive. GDP per capita may reduce fatal accidents; however, this relationship does not reflect labor related factors such as unemployment and hours worked. These findings affirm Nichols' view that the prevalence of working hours compared to unemployment indicates an impact of labor demand on workplace accidents through working conditions; his theorized lag between output and unemployment is also observed. These findings do not affirm Saloiemi and Oksanen (1998) and Boone and van Ours (2006) in that labor market variables do impact fatal injuries. These findings affirm Terres de Ercilla et al in that different variable have different effects in term of direction.

Methodology
With the exception of Song, He, and Li (2010), this paper is the most sophisticated paper. I see it as a deficiency when compared to other papers in that the better determined model in other papers didn't require so much statistical manipulation. They manipulate their equations at least twice to arrive at a statistically satisfying. I did not understand the manipulations they performed. I understand the metrics they used but not how they used them.

Current List of Articles

1. Kossoris (1938) Industrial injuries and the business cycle. Monthly Labour Review.
2. Steele (1974) Industrial Accidents: An Economic Interpretation. Applied Economics.
3. Shuster and Rhodes (1985) The Impact of Overtime Work on Industrial Accident Rates. Industrial Relations.
4. Nichols (1991) Labour Intensification, Work Injuries, and the Measurement of Percentage Utilization of Labour (PUL). British Journal of Industrial Relations.
5. Saloniemi and Oksanen (1998) Accidents and Fatal Accidents – Some Paradoxes. Safety Science.
6. Gerdtham and Johannesson (2003) A Note on the Effect of Unemployment on Mortality. Journal of Health Economics.
7. Terres de Ercilla et al. (2004) Economic fluctuations affecting occupational safety. The Spanish case. Occupational Ergonomics.
8. Boone and van Ours (2006) Are recessions good for workplace safety. Journal of Health Economics.
9. Gerdtham and Ruhm (2006) Deaths rise in good economic times: evidence from the OECD. Economics and Human Biology.
10. Song, He, and Li (2010) Longitudinal Relationship between Economic Development and Occupational Accidents in China. Accident Analysis and Prevention.
11. Mouza and Targoutzidis (2010) The Effect of the Economic Cycle on the Workplace Accidents in Six European Countries. Ege Academic Review.
12. Svensson (2010) Economic Upturns are good for Your Health but Watch Out for Accidents: A Study on Swedish Regional Data 1976 – 2005. Applied Economics.
13. Boone and van Ours (2011) Recessions are bad for Workplace Safety. Journal of Health Economics.
14. Mouza and Targoutzidis (2012) The impact of the economic cycle on fatal injuries. The case of UK 1971–2007. Quality and Quantity.